

Jisung Yoo

Senior Research Scientist | 3D Computer Vision

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PROFILE

3D vision researcher specializing in large-scale scene reconstruction, neural rendering, and generative 3D for real-world environments. At NAVER LABS, core work includes production-scale Street View 3D and city-scale novel-view synthesis deployed in NAVER Map, with contributions to aerial digital-twin reconstruction. Current research explores 3D Gaussian Splatting, generative video/3D diffusion, and feed-forward 3D representations for extending under-observed scenes while preserving geometric and multi-view consistency. Previous experience includes on-device AR, computational photography, and 3D-content service prototyping for Samsung Galaxy products.

PROFESSIONAL EXPERIENCE

NAVER LABS

Feb 2022 - Present

Senior Research Scientist | Seongnam, South Korea

- Core contributor to the large-scale 3D reconstruction and rendering pipeline behind NAVER Map Street View 3D: multi-view stereo depth reconstruction for multi-camera mapping rigs, RGB-D panoramic stitching, mesh generation, and production pipeline optimization.
- Core contributor to city-scale novel-view synthesis (NVS) for free-viewpoint urban rendering, deployed in NAVER Map Flying View 3D and media/VFX applications: dense-matching preprocessing for aerial imagery, geometry-aware tiled 3DGS training, and seamless LoD rendering across heterogeneous street-view and drone imagery.
- Contributed MVS-based dense point-cloud and mesh reconstruction, as well as 3DGS-based scene reconstruction outputs, to city-scale digital-twin pipelines.
- Current research: exploring generative video/3D diffusion and feed-forward 3D Gaussian representations to extend reconstruction and NVS beyond captured viewpoints, emphasizing geometric and multi-view consistency; a parallel thread studies appearance editing under fixed geometry.

Selected public work: [Flying View 3D - City-scale NVS](#) | [Street View 3D - Large-scale 3D Reconstruction](#) | [NVS for Media & VFX](#)

Patents: three Korean patent applications filed at NAVER LABS (2025-present), pending publication.

Samsung Research

Jan 2021 - Jan 2022

Creative Leader, Team ReMo (Reconstruct Moments) | Seoul, South Korea

- Led an incubation team exploring AR/VR applications and business models for 3D content creation from smartphone photos.
- Prototyped photorealistic 3D-content experiences, including single-image animatable avatars and interactive 3D photos from smartphone image sequences.

Samsung Electronics

Mar 2018 - Dec 2020

Staff Engineer, Mobile Communications Division | Suwon, South Korea

- Developed core time-of-flight (ToF) AR measurement technologies for QuickMeasure, commercialized on the Galaxy S10 5G.

- Developed photo-filter style transfer for My Filters on the Galaxy S20 series; corresponding author of an ECCV 2020 Spotlight paper.
- Implemented the on-device Photo Remaster pipeline for degradation detection, automatic enhancement, and result recommendation, commercialized on the Galaxy S21 series.

RESEARCH BACKGROUND

KAIST

Mar 2010 - Feb 2018

Doctoral Research, Electrical Engineering | Daejeon, South Korea

- Implemented visual SLAM and volumetric reconstruction for multi-view 3D face reconstruction, with preprocessing for eye blinks, specular regions, and keyframe selection.
- Built RGB-D and multi-view capture systems; combined visual hulls, depth refinement, and view-dependent texture mapping for 3D object reconstruction.

SELECTED PUBLICATIONS

Filter Style Transfer between Photos

J. Yim, J. Yoo*, W.-J. Do, B. Kim, and J. Choe.

*European Conference on Computer Vision (ECCV), 2020. Spotlight. *Corresponding author.*

Visual Hull-Based Geometric Data Compression of a 3-D Object

Sung Soo Hwang, Wook-Joong Kim, Jisung Yoo, and Seong Dae Kim.

IEEE Transactions on Circuits and Systems for Video Technology, 25(7):1151-1160, July 2015.

Scale-invariant template matching using histogram of dominant gradients

J. Yoo, S. S. Hwang, S. D. Kim, M. S. Ki, and J. Cha.

Pattern Recognition, 47(9):3006-3018, September 2014.

TECHNICAL EXPERTISE

Research: 3D reconstruction, multi-view stereo, RGB-D reconstruction, visual SLAM, neural rendering, novel-view synthesis, 3D Gaussian Splatting, generative video/3D models, feed-forward 3D representations.

Vision & imaging: Image processing, depth estimation, segmentation, image enhancement, style transfer.

Programming & platforms: C/C++, CUDA, OpenGL, Three.js, Android.

EDUCATION

KAIST

Mar 2010 - Feb 2018

Ph.D. in Electrical Engineering | Daejeon, South Korea

Dissertation: A study on interactive 3D reconstruction using RGB-D camera

KAIST

Mar 2006 - Jan 2010

B.Eng. in Electrical Engineering | Daejeon, South Korea

SELECTED RECOGNITION

Samsung Ph.D. Scholarship | Samsung Electronics | Mar 2013 - Feb 2017

Silver Prize, Korean Physics Olympiad (KPhO) | 2005